

MARLET: Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring

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Abstract—Large language models can generate fluent tutoring responses, but they need grounding that respects both instructional evidence and the learner’s current state. Classical retrieval-augmented generation (RAG) grounds generation through a fixed retrieve-then-read controller: the query is matched against a corpus, and the generator reads the returned passages. That design is often insufficient for tutoring, where the same surface question can require different evidence, scaffolding, or rubric constraints for different learners. We propose MARLET (Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring), which integrates supervisory routing, planning, learner-state diagnosis, rubric analysis, conditional retrieval, tutoring, and bounded critique around a fixed tutor. MARLET builds a tutoring brief from the sample, inferred learning state, and answer criteria, then invokes retrieval only when the route indicates that external evidence is needed. Under matched backbones, corpus, reranker, and response cap on TutorEval and EduBench, MARLET improves fixed-tutor quality metrics while using fewer tokens; on EduBench it also reduces latency and backbone-call count. These results support the central claim: educational grounding should be controlled by pedagogical state and task regime, not by the surface query alone.

I. Introduction

Large language models (LLMs) are increasingly used as educational tutors because they can generate explanations, diagnose errors, and provide feedback [1], [2]. However, an LLM tutor cannot rely only on parametric knowledge. Knowledge-grounded dialogue has long shown the value of external evidence [3]; educational responses similarly need grounding in textbook chapters, course notes, rubrics, worked examples, or learner profiles to avoid fluent but misaligned guidance.

Classical retrieval-augmented generation (RAG) has therefore become the default grounding mechanism [4]. In classical RAG, the query is embedded, the top- k most similar passages are retrieved, and the LLM generates a response conditioned on those passages. Thus, relevance depends heavily on retrieval quality and query representation [5].

Educational tutoring is more complex than factual question answering. A student’s question is often only a partial signal: the appropriate response may depend on level, goals, weak points, recent confusion, dialogue, and likely misconceptions. We call this the learner’s personalized learning state. For example, an ordinary-differential-equation query may require high-school procedural scaffolding or college-level modeling context.

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Therefore, tutoring retrieval must find evidence that is pedagogically appropriate for the learner, not merely topical.

Classical RAG can include explicit learner-state text when supplied, but appending learner state, dialogue, grading criteria, and task constraints forces one query embedding to compress topic, level, misconception, instructional goal, and response constraints. As constraints grow, search intent can blur. The issue is therefore how a tutoring system should organize constraints before retrieval and generation.

Therefore, we introduce MARLET (Multi-Agent Retrieval for Learning-State-Aware Educational Tutoring). Instead of letting the retriever act as the top-level controller, a supervisory router identifies the tutoring regime and decides whether retrieval is needed. A planner scopes search intent, a diagnoser summarizes visible learning state, and a rubric module states answer criteria; these modules run in parallel when possible, then are re-assembled with retrieved evidence into one tutor prompt. The framework integrates selective retrieval, tool-using agents, and tutoring-state modeling into one context-construction mechanism.

The experiments validate the framework rather than present a model leaderboard. We compare with classical retrieve-then-read RAG and Agentic RAG under shared backbone, corpus, reranker, response cap, tutor, and critic. Our validation on TutorEval and EduBench shows that the same tutor scores higher when fed context assembled by MARLET; on EduBench it also reduces latency and backbone calls.

Contributions. (i) We propose MARLET, a learning-state-aware multi-agent retrieval framework with a supervisory router, planner, diagnoser, rubric module, conditional retriever, tutor, and critic. (ii) We formalize the router and retrieval gate as transparent, training-free controls over learner state, rubric constraints, and external evidence. (iii) We validate against classical RAG and Agentic RAG baselines on two educational benchmarks under shared backbone, corpus, reranker, and response cap.

II. Related Work

Classical RAG fixes retrieval as the first controller: the query is matched to passages, and generation conditions on the returned evidence [4]. Selective and corrective RAG methods improve this controller. Self-RAG learns

when to retrieve, generate, and critique; CRAG evaluates retrieved documents and attempts correction when retrieval is unreliable; Adaptive-RAG and TARG vary retrieval behavior by query complexity or retrieval utility [6]–[9]. RAG diagnostics further emphasize separating retrieval, ranking, and generation errors [10]. These methods address whether passages retrieved for a query are necessary, sufficient, or reliable, but they do not infer learner stage before retrieval, separate rubric constraints from topic evidence, or decide whether external passages are the right source of grounding for a tutoring decision.

Agentic RAG and multi-agent LLM systems move beyond a single retrieve-then-read step. ReAct, AutoGen, CAMEL, and MetaGPT show that tool use and role decomposition can improve complex task execution [11]–[14]. MA-RAG uses multiple agents for query disambiguation, evidence extraction, and answer synthesis on ambiguous or multi-hop QA tasks [15]. These systems demonstrate coordination benefits, but their roles are designed for general task execution or ambiguous question answering. They do not define pedagogical regimes, infer visible learner state, or bind the final response to rubric-style answer criteria, so the educational reason for retrieving, skipping retrieval, or prioritizing rubric context remains implicit.

Recent educational work shows that tutoring quality requires more than answer accuracy. AgentTutor builds multi-turn personalized learning with learner assessment and memory [2]; adaptivity studies show that LLM tutors are sensitive to omitted student-context features [16]; and AI-tutor evaluation emphasizes pedagogy, mistake diagnosis, guidance quality, and learner support [1], [17]–[20]. These works define tutoring requirements, but they do not formulate retrieval as learner-state-aware control: when to retrieve, what evidence to retrieve, and how much context to allocate to evidence rather than learner-state and rubric information.

III. Methodology

MARLET turns one educational request into a grounded tutor response. It is a RAG controller rather than a set of independent answer agents: every intermediate output is converted into context for one final LLM tutor prompt. Its design integrates three principles: route before retrieval, exchange typed records rather than free-form chat, and retrieve only when the tutoring brief needs external evidence. The flow is: standardize the entry, score routing cues, run active specialists in parallel, merge their outputs with retrieved evidence, serialize one augmented prompt, and run one critic revision if needed.

A. Input Entry and Inference Caps

For one sample, MARLET first standardizes all visible fields into

$$x = (q, o, c, h, r, v). \quad (1)$$

Algorithm 1: MARLET Inference Process

Require: entry $x = (q, o, c, h, r, v)$, caps K, Q, L, N , index $\mathcal{I}$, thresholds τ .

Return: answer y and internal grounding fields $(R, D, \text{citations})$.

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1:  $b \leftarrow \text{concat}(q, c, r, h)$ ;  $s \leftarrow \text{CueScores}(b)$ 
2:  $(R, G) \leftarrow \text{Route}(s, r, v, \tau)$ 
3: parallel do
4:    $p = (\text{strategy}, P) \leftarrow \text{Planner}(x, R, G)$ 
5:    $\ell \leftarrow \text{Diagnoser}(q, h)$ 
6:   if  $r \neq \emptyset$  or  $R = \text{RF}$  then  $u \leftarrow \text{Rubric}(x, R)$  else  $u \leftarrow \text{DefaultCriteria}()$ 
7: end parallel
8:  $P \leftarrow \text{cap}(P, K)$ ;  $D \leftarrow \emptyset$ 
9: if  $G = 1$  then
10:   $D \leftarrow \text{Pack}(\text{Rerank}(\text{Dedup}(\text{Search}(\mathcal{I}, P)), q), Q, L)$ 
11: end if
12:  $y_0 \leftarrow A_{\text{tutor}}(\text{Prompt}(q, o, h, p, \ell, u, D, c), N)$ 
13: if  $D = \emptyset$  and  $s_{\text{ev}} \geq \tau_2$  then
14:   $D \leftarrow \text{Pack}(\text{Rerank}(\text{Search}(\mathcal{I}, [q]), q), Q, L)$ 
15:   $y_0 \leftarrow A_{\text{tutor}}(\text{Prompt}(q, o, h, p, \ell, u, D, c), N)$ 
16: end if
17:  $y \leftarrow \text{CriticOnce}(y_0, q, h, p, \ell, u, D, r)$ 
18: return  $y, (R, D, \text{ids}(D))$ 

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Here, q is the student question or task prompt; o is an optional answer-option list; c is optional inline context such as a chapter excerpt; h is recent dialogue; r is an optional rubric or answer-criteria list; and v is an optional image list. Empty fields are allowed, so the same representation covers factual questions, grading requests, feedback requests, and planning requests.

The only student-facing output is the final tutor answer y . MARLET also keeps internal grounding records: a route record, which names the selected tutoring mode and which modules run; an evidence bundle D , which contains packed retrieved chunks; and citation identifiers, which link chunks in D to their sources. These records are not separate answers. Planner output, learner-state summary, rubric summary, and D are assembled as prompt blocks before the LLM tutor generates y . Fixed inference caps keep the process bounded: K caps planner retrieval queries, Q caps packed chunks/tool calls, L caps packed-evidence length, and N caps response tokens.

B. Supervisor Routing

The supervisor begins with a routing blob:

$$b(x) = \text{concat}(q, c, r, h), \quad (2)$$

where $\text{concat}(\cdot)$ means that the question, inline context, rubric text, and dialogue turns are joined into one normalized string. Images remain available to the tutor, but they are not converted into the textual routing score.

The router uses four cue families $j \in \{\text{ev}, \text{ru}, \text{pl}, \text{tu}\}$ for evidence, rubric, planning, and tutoring. Each family controls a downstream decision: evidence cues open retrieval, rubric cues activate criterion checking, planning cues route to lesson sequencing, and tutoring cues activate learner-state adaptation. Each family has a keyword set W_j . Evidence cues include source, document, chapter, and verification; rubric cues include score, criteria, and comment; planning cues include lesson, sequence, and exercise plan; tutoring cues include student, hint, misconception, explain, and confused. Let $M_j(x)$ be the number of cues from W_j that appear in $b(x)$. The cue

score is

$$s_j(x) = \frac{M_j(x)}{|W_j|}, \quad (3)$$

where $|W_j|$ is the number of cues in family j , so $s_j(x)$ is the fraction of that cue family observed in the routing blob. The thresholds are fixed router hyperparameters, not probabilities or learned weights. We choose them in cue-count units so τ_{mid} requires a repeated cue pattern rather than one accidental word, τ_{plan} is stricter because a false planning route changes the whole answer structure, and τ_2 permits retry only for strong evidence cues.

After cue scoring, the supervisor chooses a pedagogical regime. A regime is not a dataset label; it is a generic prompt-priority decision made before retrieval and generation. The intuition is to ask what missing constraint would most likely make the final answer wrong. If the learner needs an ordered path, lesson planning (LP) reserves space for sequence and pacing. If the task asks for judgment, rubric feedback (RF) reserves space for criteria and scoring logic. If the learner’s background changes how the same concept should be explained, adaptive tutoring (AT) reserves space for learner state. If correctness depends mainly on outside material, evidence-grounded reasoning (EG) reserves space for source evidence. Because these needs can overlap, $R(x)$ is selected by the priority rule

$$R(x) = \begin{cases} \text{LP,} & s_{\text{pl}} \geq s_{\text{ru}}, s_{\text{pl}} \geq s_{\text{tu}}, s_{\text{pl}} \geq \tau_{\text{plan}}, \\ \text{RF,} & s_{\text{ru}} \geq s_{\text{tu}}, s_{\text{ru}} \geq \tau_{\text{mid}}, \\ \text{AT,} & s_{\text{tu}} \geq \tau_{\text{mid}}, \\ \text{EG,} & \text{otherwise.} \end{cases} \quad (4)$$

The order reflects this generic constraint strength. Sequence changes the whole response, explicit criteria constrain evaluation, learner state shapes delivery, and evidence is the default grounding route when the request does not clearly require one of the first three prompt shapes.

Next, the supervisor uses a retrieval gate: a binary control for whether external evidence enters the final tutor prompt. Inspired by selective-retrieval research, MARLET integrates this idea as a prompt-allocation control that opens only when outside grounding is needed [6], [8], [9]:

$$G(x) = \begin{cases} 1, & s_{\text{ev}} \geq \tau_{\text{mid}} \text{ or } R(x) = \text{EG,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Thus $G(x) = 1$ runs retrieval; $G(x) = 0$ builds the same prompt without external chunks. A stricter retry gate runs one raw-question pass only when the first bundle is empty:

$$G_2(x) = \begin{cases} 1, & D_0 \text{ is empty and } s_{\text{ev}} \geq \tau_2, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

where D_0 is the evidence bundle from the first retrieval attempt.

C. Parallel Agent Brief Construction

The router and gates above decide what the final prompt should prioritize; MARLET then constructs the prompt content as a shared tutoring brief. Inspired by tool-using and multi-agent LLM systems [11], [12], planning chooses the local answer strategy and search intents, while coordination means typed shared-state assembly: specialists read the same input, write their own fields, and a coordinator merges them before retrieval and generation. The planner, diagnoser, and rubric module run in parallel:

$$(p, \ell, u) = (A_{\text{plan}}(x, R, G), A_{\text{diag}}(q, h), A_{\text{rub}}(x, R)). \quad (7)$$

Here p is the planner output, consisting of a response strategy and up to K retrieval queries; ℓ is the visible learner-state summary; and u is the rubric summary or default checklist. In parallel execution, the agents read the same standardized entry and route, then return typed outputs independently, so the tutor need not infer plan, learner state, and rubric from one overloaded query.

The planner starts its query set with q , adds a compact key-term query from longer tokens in q , adds a context-aware query when c is present, and adds a rubric-aware query when r is present. Duplicate queries are removed and the list is capped by K . The diagnoser reflects tutoring work showing that LLM tutor quality depends on student-context features [2], [16]. It infers only visible state from q and h : level defaults to intermediate, beginner or advanced cues can override it, dialogue cues can add misconceptions, and style cues can request step-by-step, concise, or analogy-based explanation. This is not a persistent student profile. The rubric module preserves explicit criteria when available; otherwise it supplies default criteria of correctness, clarity, scaffolding, and actionable feedback. The merged tutoring brief is the typed record (R, G, p, ℓ, u) .

D. Conditional Retrieval and Evidence Packing

MARLET contains retrieval, but retrieval is not the top-level controller and does not require a traditional dense vector database. When $G(x) = 1$, each planner query q_i searches a local hybrid text index. For a query q_i and candidate chunk d , the base retrieval score is

$$\text{base}(q_i, d) = w_{\text{lex}} \text{lex} + w_{\text{char}} \text{char} + w_{\text{svd}} \text{svd} + \beta. \quad (8)$$

Here lex is word TF-IDF similarity, char is character 3–5 gram TF-IDF similarity, svd is truncated-SVD similarity, β is a small title/source boost, and the w terms are fixed retriever weights. Candidates from all planner queries are merged, and duplicate chunks keep their highest score.

The reranker then scores each candidate against the original question:

$$\text{rank}(q, d) = \alpha_{\text{base}} \text{base} + \alpha_{\text{tok}} o_{\text{tok}} + \alpha_{\text{title}} o_{\text{title}} + \alpha_{\text{phr}} m_{\text{phr}} + \alpha_{\text{num}} o_{\text{num}} + \alpha_{\text{len}} b_{\text{len}}. \quad (9)$$

Here o_{tok} is token overlap with q , o_{title} is title-token overlap, m_{phr} is an exact phrase-match flag, o_{num} is numeric

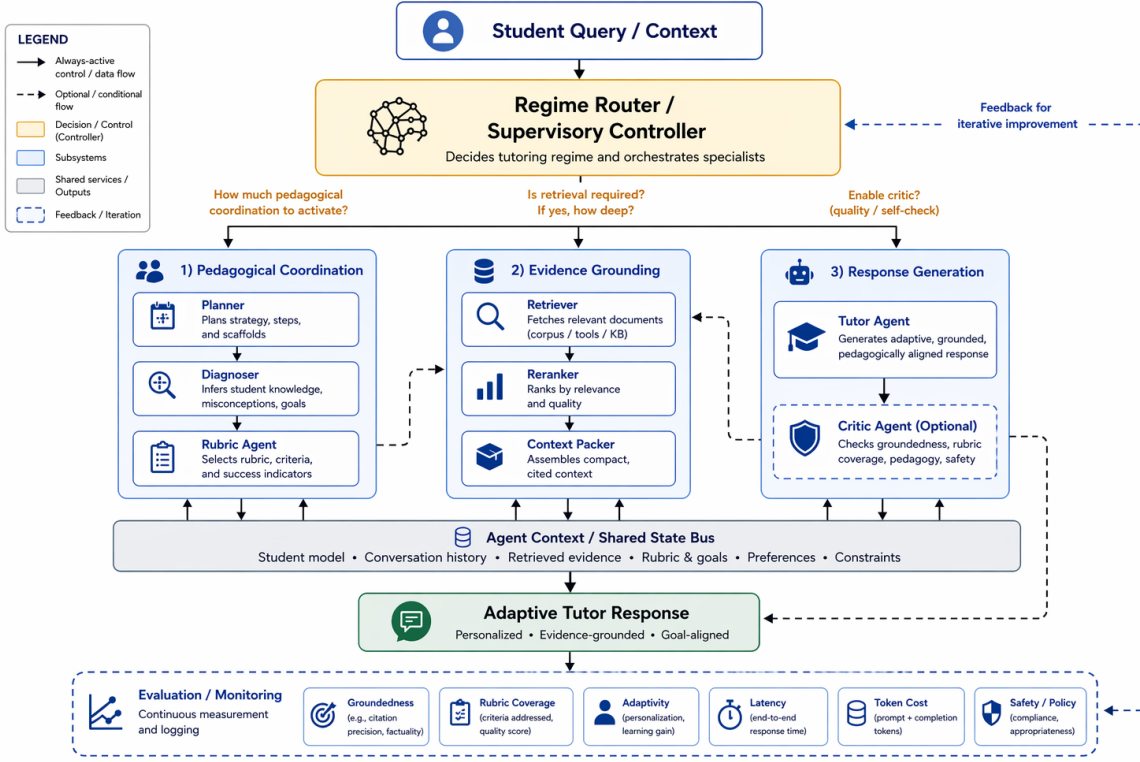


Fig. 1. Main framework of MARLET. The router scores the entry, activates specialist modules, and decides whether retrieval is needed; specialist outputs and retrieved evidence are then assembled into one augmented tutor prompt.

overlap, b_{len} rewards moderate chunk length, and the α terms are fixed reranker weights. The learner summary ℓ affects retrieval through the route and planner queries, not these weights.

Evidence packing integrates the relevance-diversity tradeoff used in MMR-style selection [21]: the context should be useful, but repeated chunks should not consume the limited evidence space. For each candidate chunk d , MARLET computes

$$\text{pack}(d) = \lambda \text{rank}(q, d) - (1 - \lambda) \text{repeat}(d). \quad (10)$$

Here $\text{repeat}(d)$ is the largest sentence overlap between d and selected chunks, and λ controls the relevance-diversity balance. The packer adds the remaining chunk with the highest $\text{pack}(d)$ score until Q chunks or the L character limit is reached, forming evidence bundle D .

E. Tutor Generation and Critic Revision

This is where retrieval and agent outputs converge: (R, G, p, ℓ, u) supplies route, plan, learner-state, and rubric fields, while D supplies compact cited evidence. The retriever does not answer; its chunks become an evidence block. The final prompt contains task fields, pedagogical brief, evidence or inline context, and output requirements. The tutor is the LLM-driven answer writer:

$$y_0 = A_{\text{tutor}}(q, o, h, p, \ell, u, D, c). \quad (11)$$

Its prompt contains q , optional answer options o , recent dialogue h , learner state ℓ , plan p , rubric summary u , and

either retrieved evidence D or inline context c . The tutor is instructed to be correct, pedagogically useful, concise but not terse, include a next step when appropriate, and cite retrieved evidence with document markers.

The critic performs one bounded revision pass. It checks for missing citation markers when D is nonempty, uncovered rubric items from r , or an answer that is too long for a beginner-facing state. If no issue is found, $y = y_0$; otherwise the critic rewrites once using $(y_0, q, h, p, \ell, u, D, r)$ and the revised y becomes the final answer.

IV. Validation Experiments

We validate the framework on two public educational benchmarks. TutorEval [1] contains 834 expert-written science tutoring questions paired with long STEM chapters and teaching key points, so responses must explain, disambiguate, and stay chapter-grounded. EduBench [18] covers nine educational scenarios, including question answering, error correction, personalized support, emotional support, grading, teaching-material generation, and personalized content creation. Its rows contain prompts, scenario metadata, and reference model predictions; our adapter converts them into consensus-style supervision through score statistics and rubric terms.

The primary validation compares three context controllers under matched infrastructure, with a Qwen3.5-72B-FP8 replication checking backbone sensitivity. Clas-

sical RAG retrieves first from the raw standardized prompt; visible learner profile, student answer, dialogue, and rubric text remain available to its tutor prompt, so the baseline is not deprived of pedagogical context. Its limitation is controller order: retrieval happens before specialist summaries can shape search. Agentic RAG runs the same planner, diagnoser, and rubric modules before retrieval, but retrieval is always on. MARLET keeps the same specialists and retriever, then adds supervisory routing, conditional retrieval, and retry control. We report quality and cost together because only context construction changes.

A. Implementation Details

All controllers use the same deployed Python framework, corpus, hybrid text index, reranker, tutor, critic, and metric code; only context assembly differs. Runs use Qwen3.5-4B and Qwen3.5-27B-FP8, reasoning budget 512, temperature 0.15, and response cap $N=900$. The checked server benwulab-Lambda-Vector runs Ubuntu Linux 6.8 and Python 3.10.12 with two NVIDIA RTX 6000 Ada GPUs (49140 MiB each), driver 570.207, and CUDA 12.8. We set $K=3$, $Q=4$, and $L=6000$; router thresholds $(\tau_{\text{mid}}, \tau_{\text{plan}}, \tau_2) = (0.35, 0.42, 0.45)$; retriever weights $(w_{\text{lex}}, w_{\text{char}}, w_{\text{svd}}) = (0.58, 0.17, 0.25)$; reranker weights $(\alpha_{\text{base}}, \alpha_{\text{tok}}, \alpha_{\text{title}}, \alpha_{\text{phr}}, \alpha_{\text{num}}, \alpha_{\text{len}}) = (0.55, 0.20, 0.10, 0.06, 0.05, 0.04)$; and packing weight $\lambda=0.68$.

Evaluation is dataset-aware, and all reported quality scores are automatic proxies rather than official benchmark or human grades. Higher is better for quality metrics; lower is better for latency, tokens, and calls. Inspired by RAG evaluation work [10], we separate answer overlap, rubric fulfillment, grounding, and resource cost. Token-F1 rewards overlap with the reference while balancing extra generated tokens against missed reference tokens. Rubric coverage checks whether each benchmark rubric item is actually expressed; near-verbatim coverage receives credit, and partial token overlap must include content-bearing words so that generic feedback is not over-rewarded. Latency is wall-clock time per sample; total tokens are prompt plus completion tokens; backbone calls count logged LLM requests.

For EduBench, EB12D averages 12 normalized checks in three groups: scenario adaptation, factual or reasoning behavior, and pedagogical application. These checks cover output-format compliance, supportive tone, relevance, use of scenario details, score agreement, domain and rubric grounding, reasoning quality, correction cues, clarity, motivation, personalization, and higher-order prompting. EduAlign is narrower: it only measures how close the extracted numeric score is to the benchmark’s consensus score. Thus EduAlign can decrease even when explanatory feedback or rubric coverage improves. For TutorEval, tutor-hit measures expert teaching-point coverage. A key point receives full credit when it is explicitly stated or strongly covered, partial credit when the answer

TABLE I
4B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.112	0.115	+0.004*
	TE hit	0.938	0.957	+0.019**
	Rubric	0.919	0.944	+0.026**
	Latency (s)	51.90	49.11	-2.79
	Tokens	3975	3490	-484***
	Calls	1.84	1.56	-0.28
EduBench	EB12D	0.367	0.398	+0.031***
	Rubric	0.705	0.891	+0.186***
	EduAlign	0.166	0.126	-0.040**
	Latency (s)	19.83	13.90	-5.92***
	Tokens	4020	2226	-1794***
	Calls	1.97	1.48	-0.49***

*** $p<10^{-4}$, ** $p<0.01$, * $p<0.05$. Higher is better for quality; lower is better for latency, tokens, and calls. Runtime is mean wall-clock seconds/sample on the experiment server.

TABLE II
27B quality and resource summary.

Data	Metric	RAG	Ours	Δ
TutorEval	Token-F1	0.264	0.272	+0.008*
	TE hit	0.897	0.918	+0.021*
	Rubric	0.861	0.891	+0.030*
	Latency (s)	41.47	40.65	-0.82**
	Tokens	6681	6730	+49
	Calls	1.99	1.99	-0.01
EduBench	EB12D	0.355	0.425	+0.070***
	Rubric	0.634	0.769	+0.135***
	EduAlign	0.541	0.526	-0.015
	Latency (s)	41.91	32.09	-9.81***
	Tokens	6105	3717	-2388***
	Calls	2.00	1.81	-0.19***

TutorEval uses $n=400$ paired samples; EduBench uses $n=328$ unique paired rows after duplicate public IDs in the 400-example slice are collapsed. *** $p<10^{-4}$, ** $p<0.01$, * $p<0.05$. Higher is better for quality; lower is better for latency, tokens, and calls. Runtime is mean wall-clock seconds/sample on the experiment server.

overlaps with enough of the key point, and no credit when the idea is absent. Off-profile metrics are omitted from paper tables. The workload summary is used only for efficiency inspection and is not a training objective.

B. Framework Validation

Table I reports paired 4B comparisons with bootstrap CIs and permutation p-values. Agentic RAG uses the same specialist brief as MARLET but retrieves for every sample; its matched results are being generated under the same protocol. On TutorEval, MARLET raises Token-F1, tutor-hit, and rubric coverage while using fewer tokens and fewer calls; latency trends lower but is not reliable ($p=0.21$). Table II reports the 27B replication.

C. Trade-offs

For TutorEval, the 27B paired CIs are positive for Token-F1, tutor-hit, rubric coverage, and latency. For EduBench, EB12D and rubric coverage improve, latency/tokens/calls decrease, and the small classical-RAG EduAlign edge is not statistically reliable.

MARLET can use more modules than classical RAG, so resource cost must be reported. Scoped planner queries and the retrieval gate reduce token use in 4B and still reduce latency in 27B, but 27B TutorEval shows that better coverage does not always mean fewer

tokens. Thus MARLET is a controller for deciding what grounding a tutoring sample needs, not a universal retrieval replacement.

V. Limitations

The study validates a framework rather than a deployable tutor: evaluation is automatic and English-only, and learner state is inferred from visible prompt/dialogue fields. Deployment requires privacy safeguards, bias checks, data governance, human oversight, and learning-gain studies.

VI. Conclusion

MARLET reframes educational grounding as supervised context construction rather than query-only passage lookup. With shared infrastructure, it improves fixed-tutor quality and efficiency over classical RAG by coordinating learner state, rubric constraints, and conditional evidence before generation. Future work should test stronger controller baselines, human and multi-turn outcomes, learned router calibration, profile-aware reranking, and privacy-preserving learner-state inference.

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